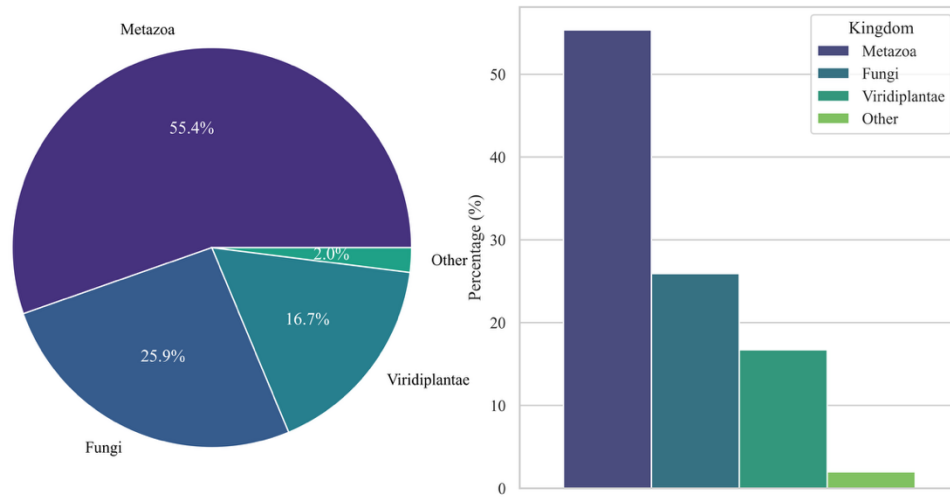
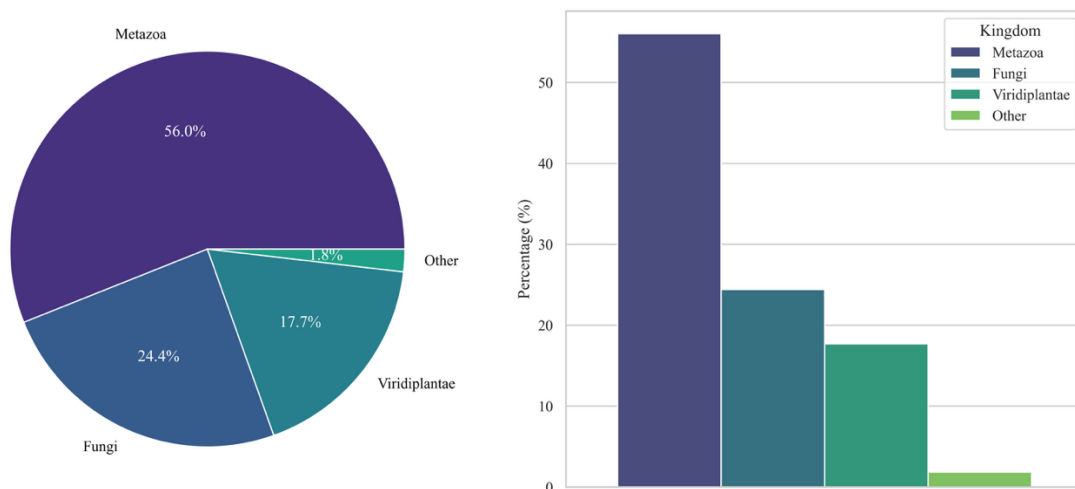


## Supplementary Materials

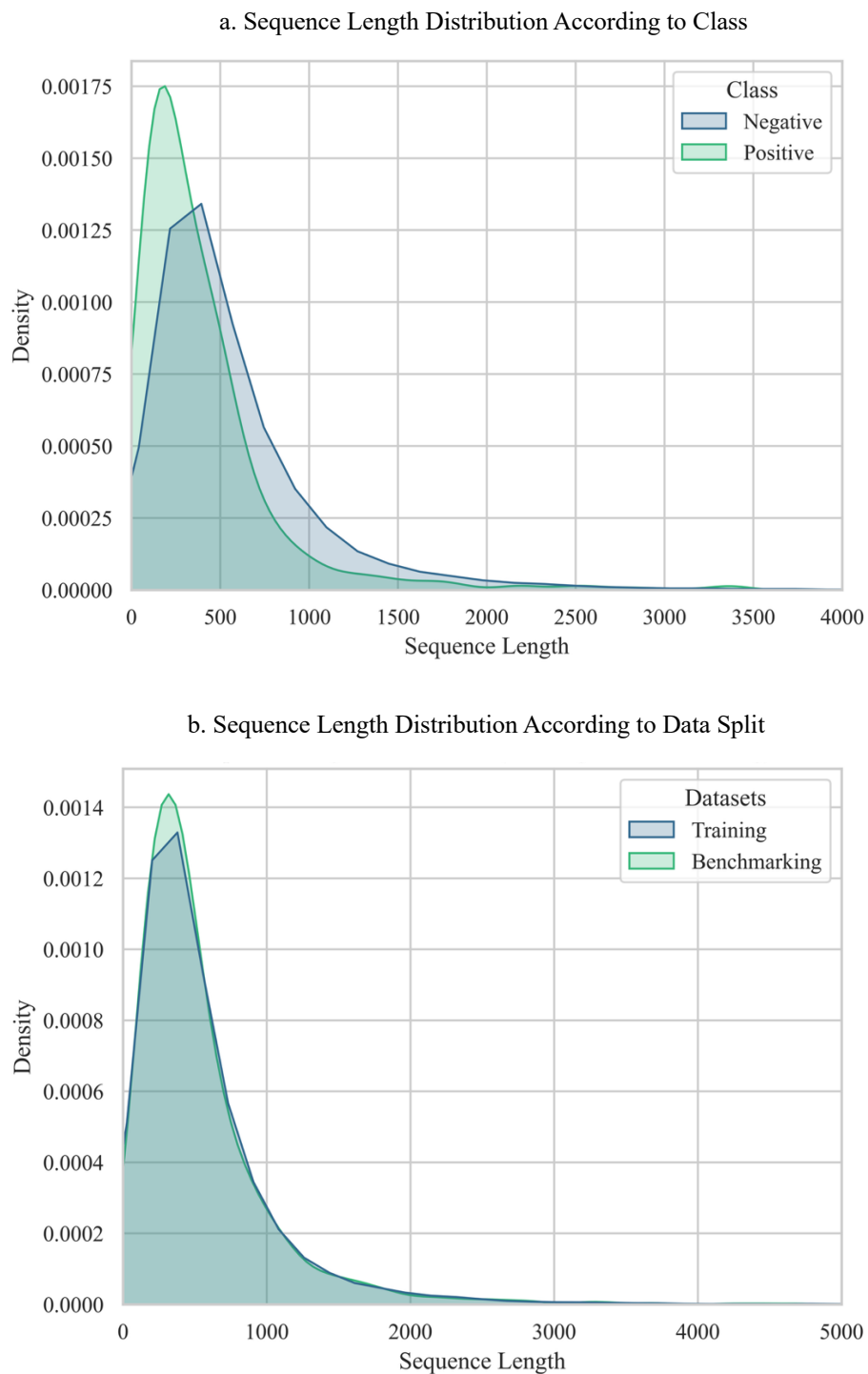
a. Training Set



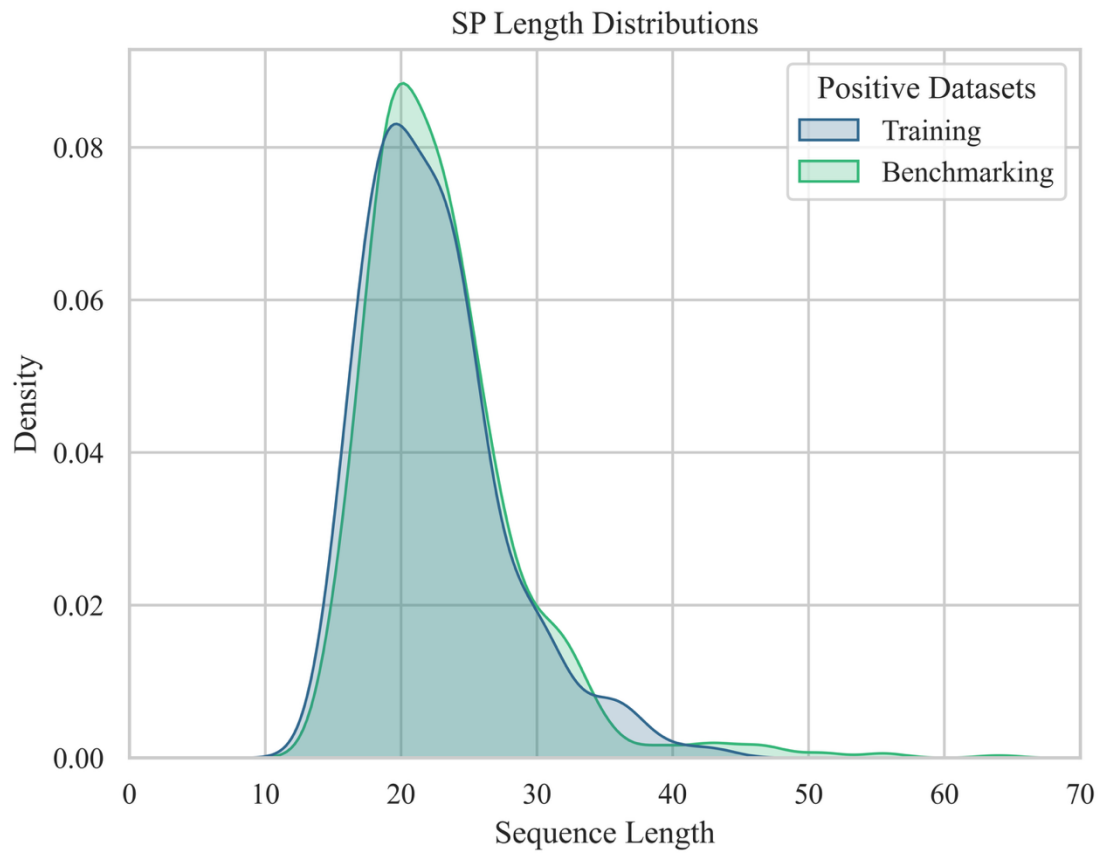
b. Benchmarking Set



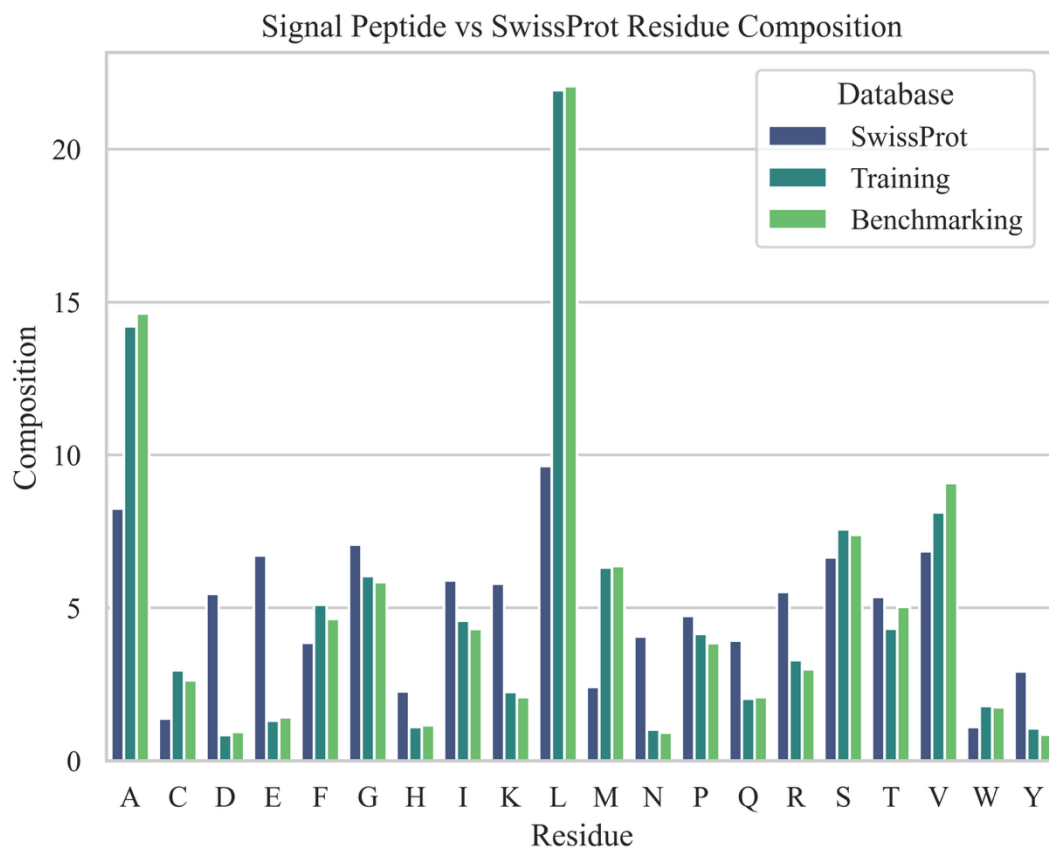
**Figure S1.** Kingdom distribution across the datasets. (a) Training set and (b) benchmarking set.



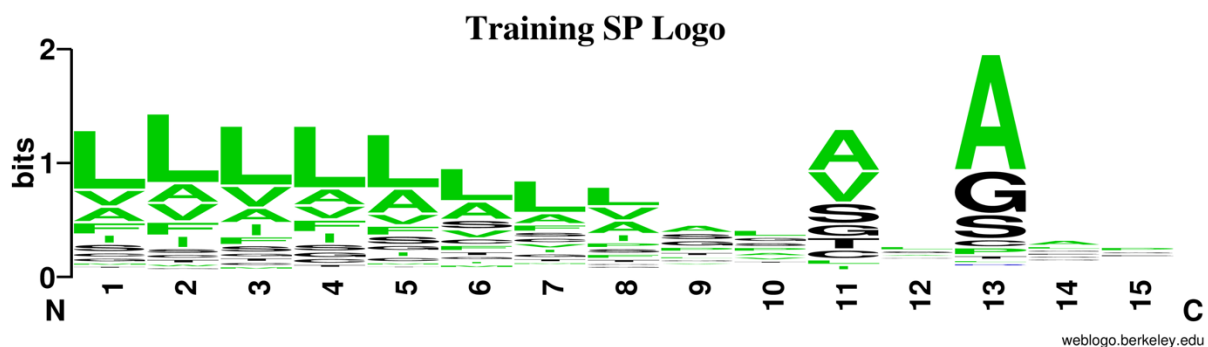
**Figure S2.** Sequence length distribution. (a) According to Class. In blue the negatives, in light green the positives. (b) According to Data Split. In blue the training set and in light green the benchmarking set.



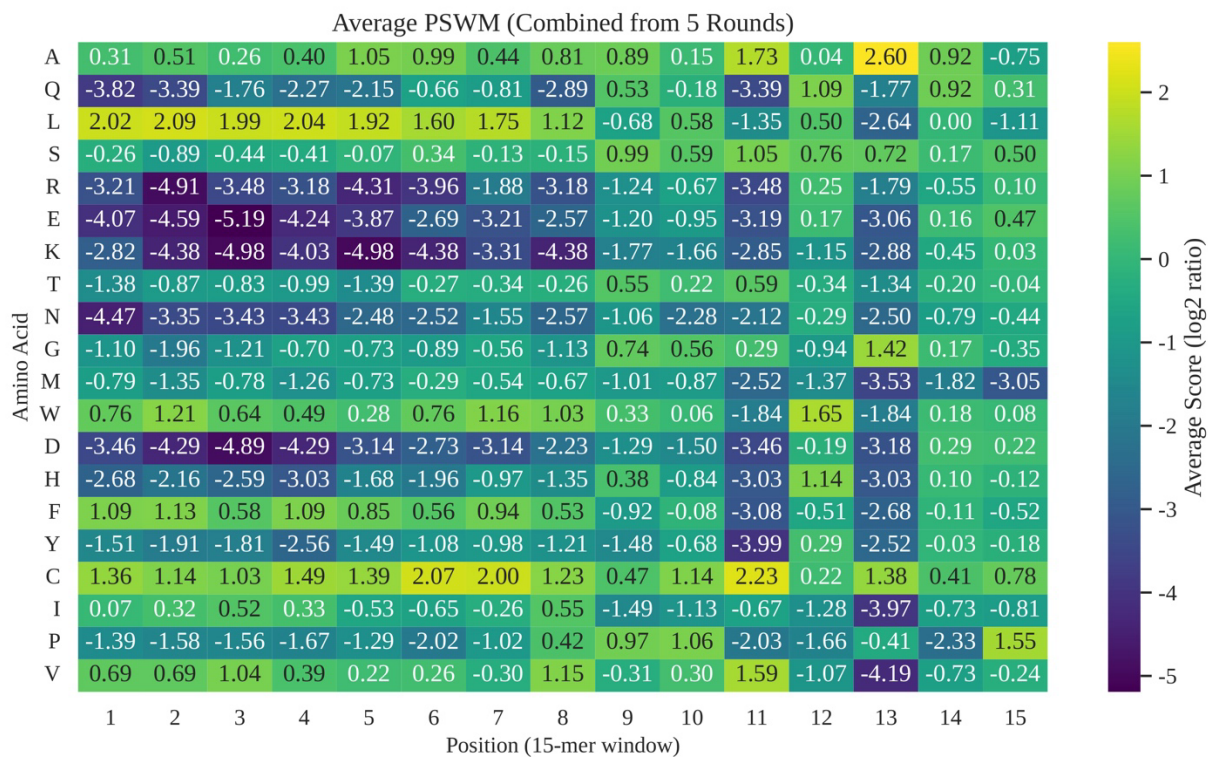
**Figure S3.** SP length distribution. In blue the training set and in light green the benchmarking set.



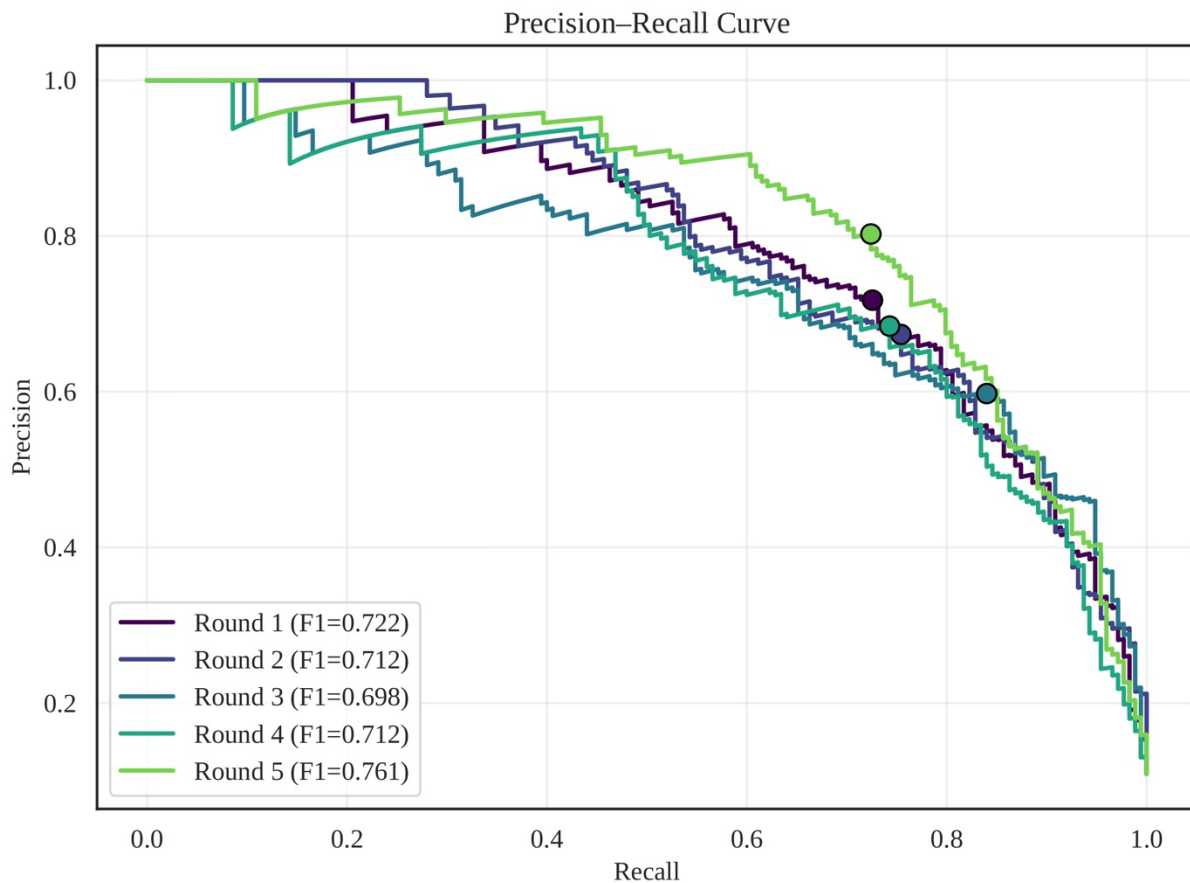
**Figure S4.** Comparison of amino acid frequencies in signal peptides from the training set (turquoise), benchmarking set (green), and the UniProtKB/Swiss-Prot reference composition (blue). Apolar residues, particularly alanine and leucine, are strongly overrepresented in both SP datasets (with leucine showing approximately 2-fold enrichment), whereas charged residues are depleted relative to the Swiss-Prot baseline. These trends are consistent across training and benchmarking sequences and reflect canonical SP physicochemical properties.



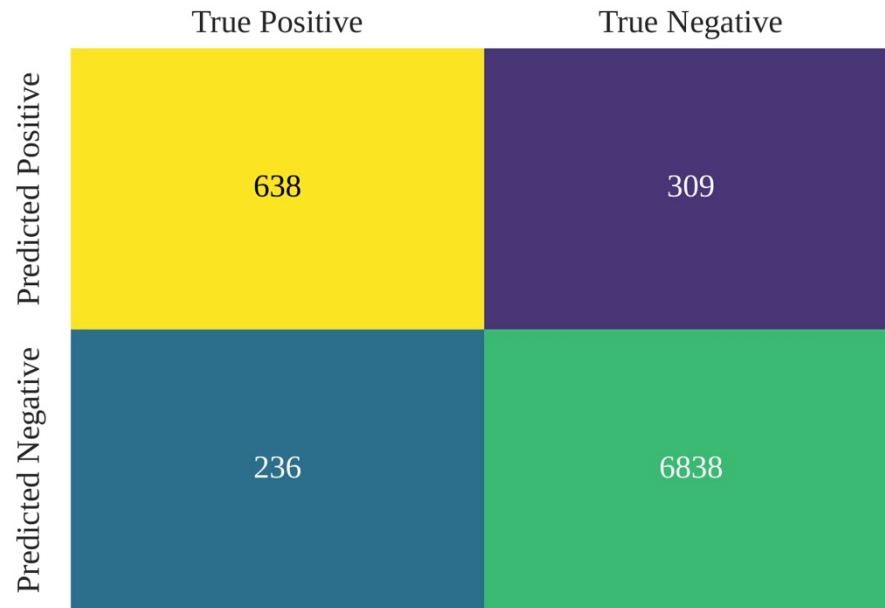
**Figure S5.** Cleavage-site sequence logo for training-set signal peptides. Sequence logo generated from 15-residue motifs (13 positions upstream and 2 downstream of the cleavage site) in the training dataset. The canonical A-X-A motif is clearly observable at the cleavage position.



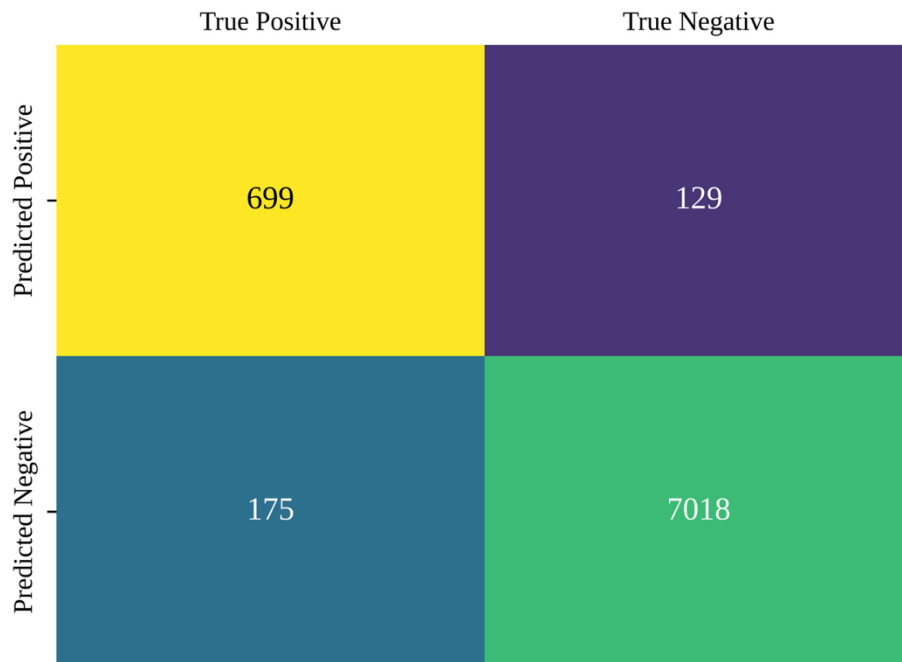
**Figure S6.** Averaged Position-Specific Weight Matrix (PSWM) derived from the five cross-validation rounds. Positive values (yellow to light green) indicate residues enriched at a given position relative to their background frequencies in Swiss-Prot, whereas negative values (blue to purple) indicate depletion.



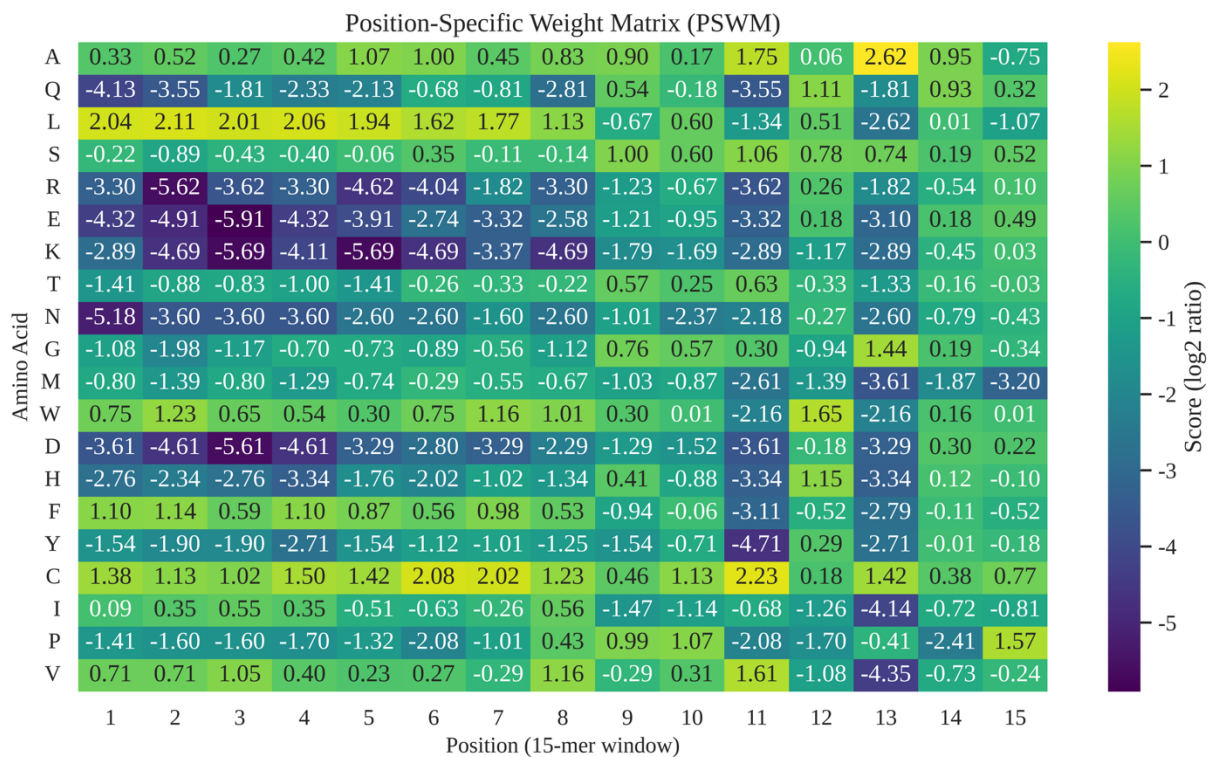
**Figure S7.** Precision–Recall curves across the five cross-validation rounds for the von Heijne classifier. Curves display broadly consistent behavior across rounds. The optimal operating point for each round, defined as the threshold maximizing the F1-score, is marked with a filled circle and annotated with the corresponding F1 value. Overall, the curves highlight stable predictive performance under class imbalance and illustrate the variability in precision–recall trade-offs inherent to the dataset.



**Figure S8.** Confusion matrix averaged on the five cross-validation rounds for the von Heijne method.

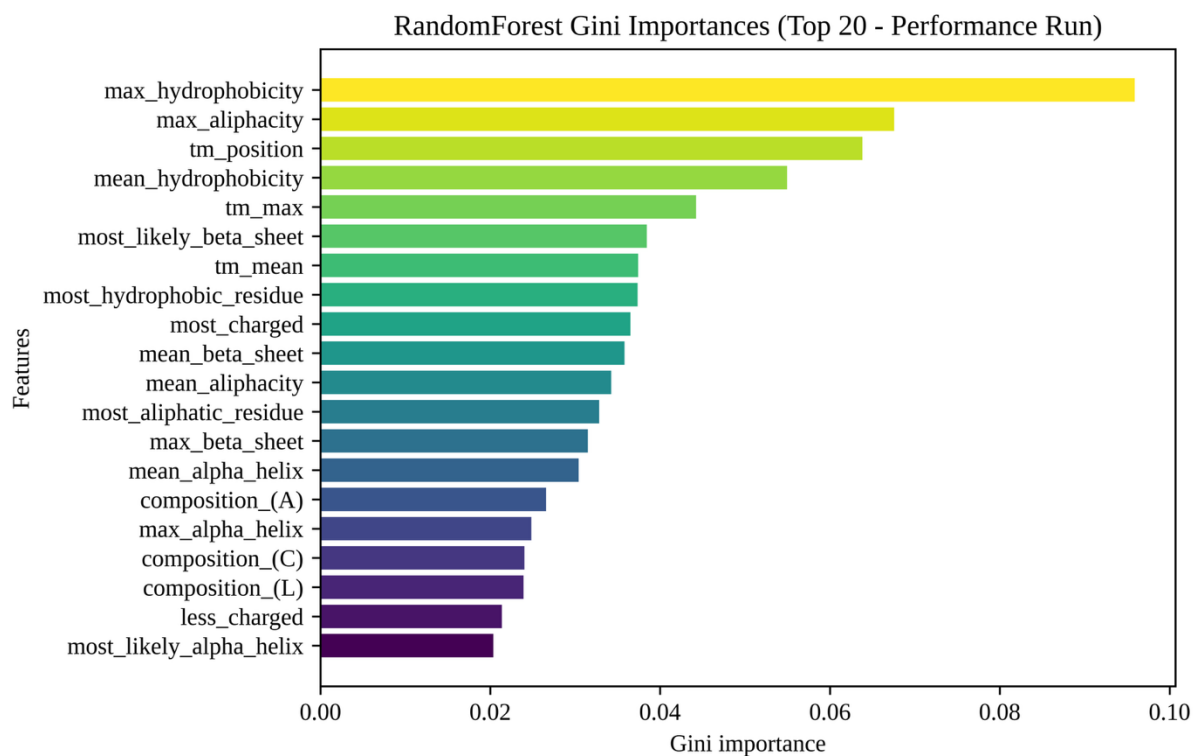


**Figure S9.** Confusion matrix averaged on the five cross-validation rounds for the SVM model.

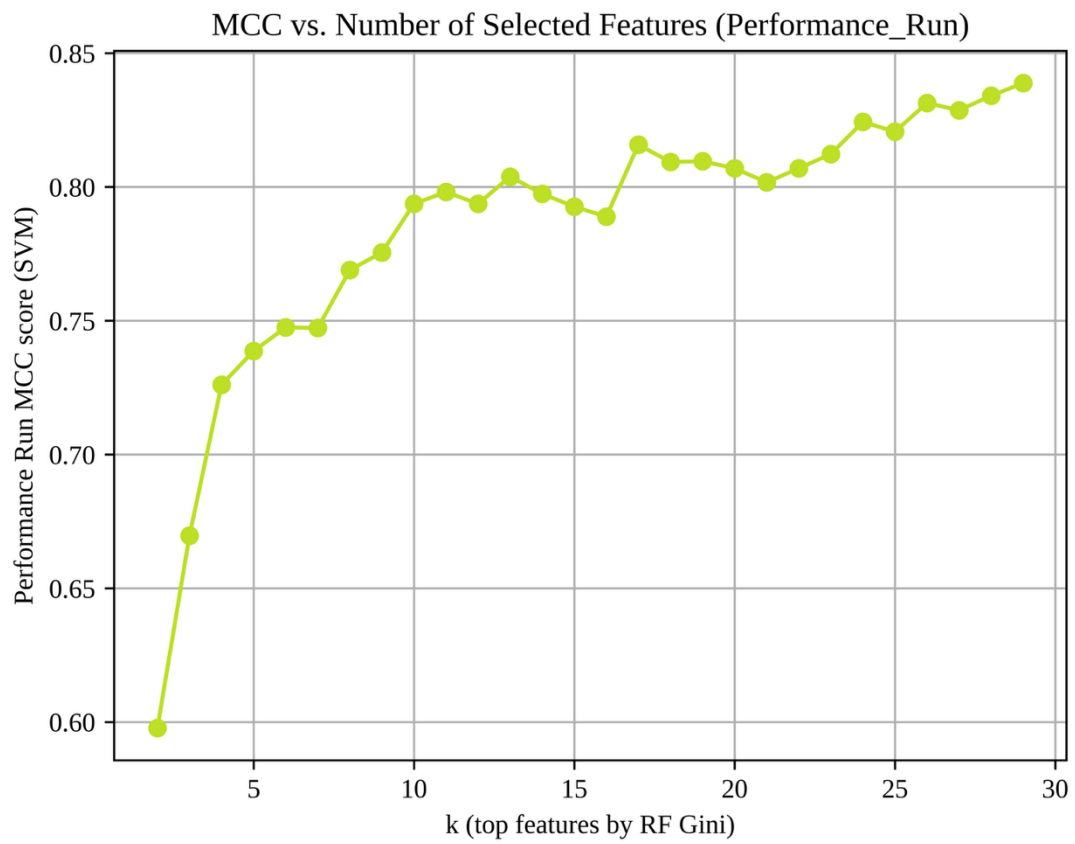


**Figure S10.** PSWM derived from the whole training dataset. Positive values (yellow to light green) indicate residues enriched at a given position relative to their background frequencies in Swiss-Prot, whereas negative values (blue to purple) indicate depletion.

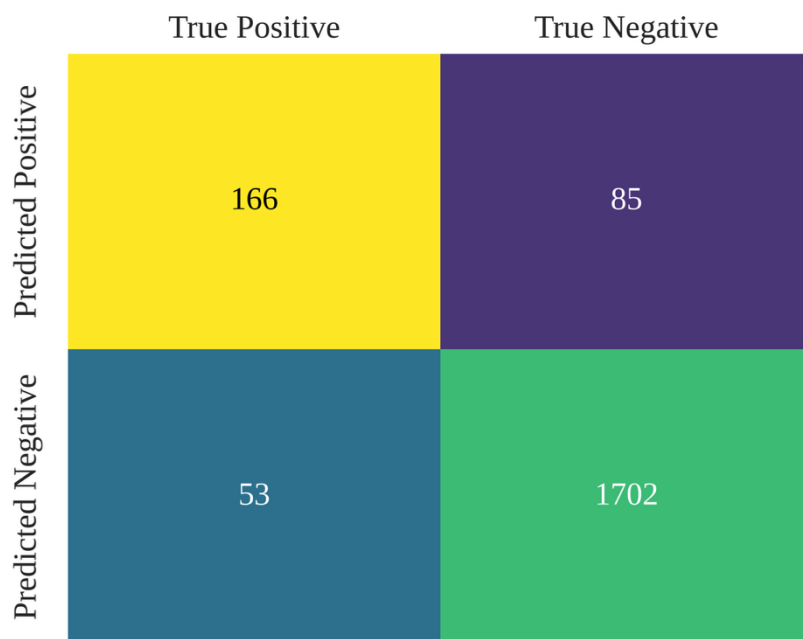




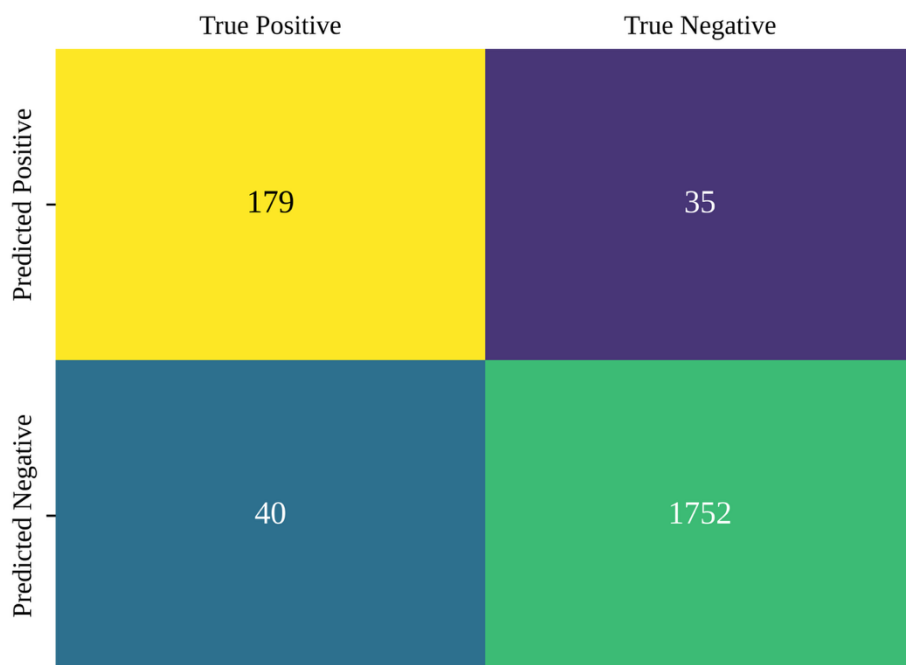
**Figure S11.** Top 20 benchmarking features ranked with respect to Gini importances in descending order.



**Figure S12.** MCC performance in the framework of the benchmarking evaluation procedure as a function of the number of selected features for the SVM classifier.

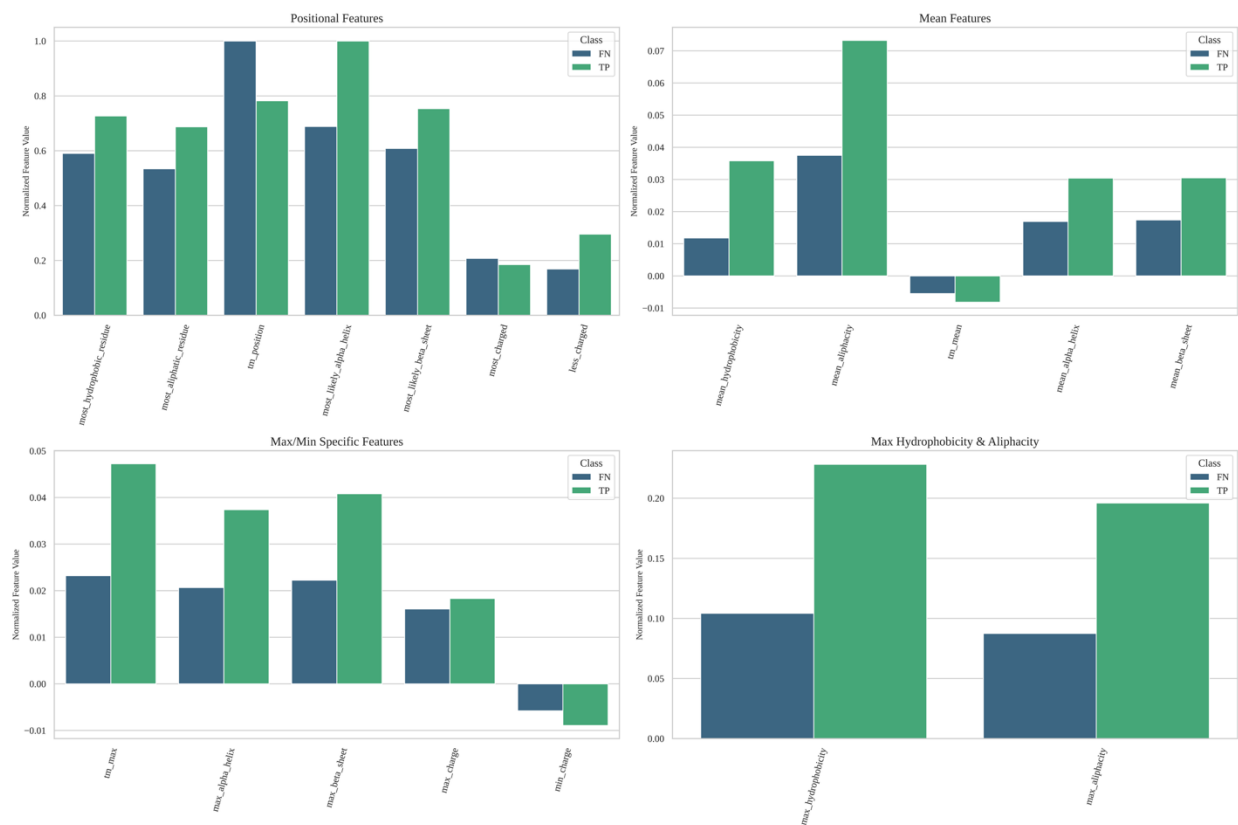


**Figure S13.** Confusion matrix derived from benchmarking procedure following the von Heijne method pipeline.



**Figure S14.** Confusion matrix derived from benchmarking procedure following the SVM selection pipeline.

Feature Comparisons: TP vs FN



**Figure S15.** Feature trends for FN and TP resulting from the SVM benchmarking evaluation procedure.